

0 8

Scientists investigated the production of laboratory rats with the characteristics of type II diabetes. The scientists used the following method.

- They divided the rats into two groups, **A** and **B**, and fed them different diets for 2 weeks.
- They fed the rats in group **A** the normal diet containing 12% fat.
- They fed the rats in group **B** a high-fat diet containing 56% fat.
- After 2 weeks, they injected both groups of rats with  $35 \text{ mg kg}^{-1}$  of the drug streptozotocin (STZ) to induce diabetes.
- 1 week later, the scientists determined the mean body mass and mean blood glucose concentration for each group.

**Table 3** shows the results.

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A value of  $\pm 2 \times \text{SD}$  from the mean includes over 95% of the data.

| Group    | Mean body mass / g ( $\pm 2 \times \text{SD}$ ) | Mean blood glucose concentration / $\text{mg dm}^{-3}$ ( $\pm 2 \times \text{SD}$ ) |
|----------|-------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>A</b> | 221.07 ( $\pm 3.28$ )                           | 129.41 ( $\pm 8.34$ )                                                               |
| <b>B</b> | 233.34 ( $\pm 5.73$ )                           | 385.02 ( $\pm 7.75$ )                                                               |

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Calculate how many grams of STZ should be injected into a rat with a mass of 230.45 g. Show your working.

Give your answer in standard form.

**[2 marks]**

Answer \_\_\_\_\_ g

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Suggest and explain why STZ was injected per unit of body mass.

**[1 mark]**

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Use all the information and your knowledge of type II diabetes to evaluate this conclusion.

**[5 marks]**

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The scientists repeated the investigation using much higher doses of STZ. This led to destruction of pancreatic cells. The scientists concluded that these rats would **not** be suitable for studying type II diabetes.

Give **two** reasons why the scientists made this conclusion.

[2 marks]

1 \_\_\_\_\_

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2 \_\_\_\_\_

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1

Turn over for the next question

